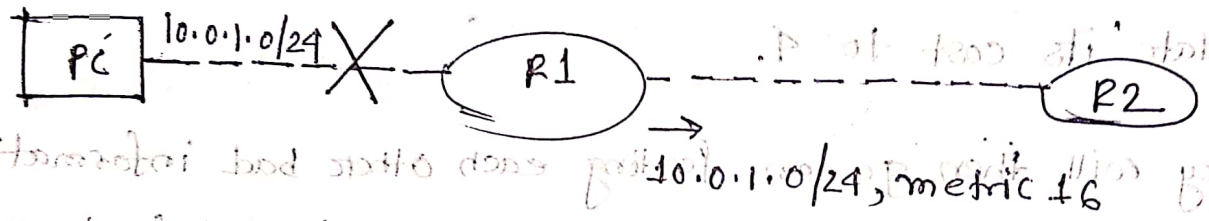
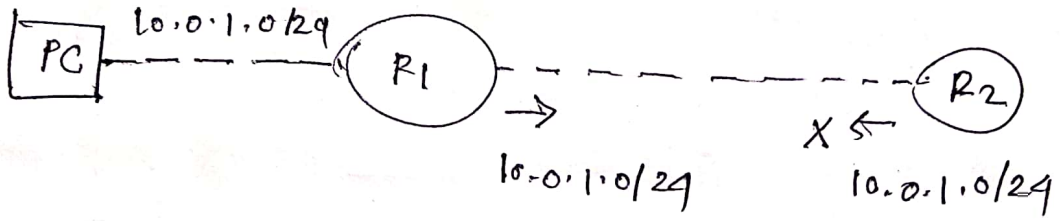
Solution :

1) Route Poisoning : When a route fails, distance vector protocols spread the bad news about a route failure by poisoning the route. Route Poisoning refers to the practice of advertising a route, but with a special metric value called Infinity. Each distance vector routing protocol uses the concept of an actual metric value that represents Infinity. RIP defines infinity as 16.



2) Split horizon :



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9. Short notes :

RIP (Routing information Protocol) :

It is a dynamic routing Protocol which uses hop count as a routing metric to find the best path between the source and the destination network. It is a distance vector routing protocol which has AD value 120 and works on the application layer of OSI model.

Features :

1. Updates of the networks are exchanged Periodically.
2. Updates are always broadcast.
3. Full routing tables are sent in update.
4. Routers always trust on routing information recieved from neighbor routers.

Advantage :

1. Easy to understand & configure.
2. Support load balancing
3. Generally loop free.
4. Almost gurranted to be supported by all routers.

Disadvantage :

1. Inefficient
2. Slow convergence in large networks.
3. Pinhole congestion is a problem.
4. Limited scalability

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BGP (Border Gate Protocol): BGP is a routing protocol used to transfer data and information between different host gateways, the internet or autonomous systems. BGP is a path vector protocol, which maintains path to different hosts. It doesn't use Interior Gateway protocol (IGP) metrics for routing decision.

Features:

1. Provide communication between two autonomous system.
2. BGP support Next-Hop algorithm.
3. Path information
4. Policy support.
5. Runs over TCP
6. Support CIDR, security

Advantages:

1. High effeiency
2. Good for ISPs
3. Prevent loops when there are multiple physical link
4. Load balance over redundant links.

Disadvantages:

1. Not stable
2. No digital identifier
3. Can create loop
4. If misconfigured, chaos can ensue by advertising itself.

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EIGP (Enhanced Interior Gateway Protocol):

It is a dynamic routing protocol which is used to find the best path between any two layer 3 device to deliver the packet. EIGP works on network layer protocol of OSI model.

Features:

1. Send hello packet every five seconds to neighbors to see if the neighbors are still available.
2. updates other routers by notifying them only when routes change.

Advantages:

1. Support IP and IPX.
2. Requires less CPU than OSPF
3. Requires little bandwith for routing updates.
4. Supports VLSM or CIDR
5. Uses the delay, bandwith, reliability of a link as its metric

Disadvantage:

1. All routers must be from Cisco systems.

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OSPF (Open Short Path First) :

It is a link-state routing protocol which is used to find the best path between the source and the destination routers using its own shortest path first.

Features :

1. Periodically sending hello packets to neighbors
2. flooding changes to neighbors when a link's status change

Advantages :

1. Routing update packets are small.
2. Not prone to routing loop
3. Scales very well to large networks.
4. Support VLSM & CIDR

Disadvantage :

1. More complex to configure & understand than a distance vector protocol.

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EGIP (Exterior Gateway Protocol) :

It is a protocol for exchanging routing information between two neighbor gateway hosts in a network of autonomous systems. EGIP is used between hosts on the Internet to exchange routing table information.

IGIP (Interior Gateway Protocol) :

A Protocol for exchanging routing information between gateways within an autonomous network. The routing information can then be used by Internet Protocol. Two commonly used IGPs : RIP & OSPF.

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